FUNDAMENTALS OF CYBER SECURITY
MODULE 6 — HANDWRITTEN QUESTIONS & ANSWERS

Exam-ready notes prepared from all 20 questions listed under Module 6 in the uploaded FCSDF Question Bank. Legal references are explained in the context and terminology used by the question bank.

Q1. Explain the importance of Cyber Security Regulations in protecting digital infrastructure.

Answer:

Cyber Security Regulations are laws, rules and official requirements that guide how organizations and individuals protect digital systems, networks and information.

Importance:

1. Protect critical infrastructure: Help safeguard sectors such as banking, energy, healthcare, transport and communication.
2. Protect personal information: Establish requirements for handling and securing sensitive data.
3. Prevent cybercrime: Define prohibited activities and legal consequences.
4. Improve organizational security: Encourage risk assessments, security controls and incident-response planning.
5. Ensure accountability: Clarify the responsibilities of organizations and service providers.
6. Support incident reporting: Help authorities respond to cyber incidents and investigate attacks.
7. Build trust: Increase public confidence in digital services and online transactions.
8. Encourage cooperation: Support information sharing between government, industry and other stakeholders.

Example:

A bank follows applicable cybersecurity requirements by using access controls, encryption, security monitoring and incident-response procedures to protect customer information.

Conclusion: Cybersecurity regulations create a legal and organizational framework that helps protect digital infrastructure and improve trust in cyberspace.

Q2. Describe the role of International Law in dealing with cross-border cybercrimes.

Answer:

International Law provides principles, agreements and cooperation mechanisms that help countries address cybercrime when offenders, victims, servers or digital evidence are located in different countries.

Role of International Law:

1. International cooperation: Encourages countries to work together in cybercrime investigations.
2. Evidence sharing: Provides legal procedures for requesting electronic evidence from another country.
3. Extradition and mutual assistance: Supports lawful cooperation in locating suspects and obtaining evidence.
4. Common standards: Encourages countries to adopt compatible cybercrime laws and procedures.
5. State responsibility: Provides principles for considering the responsibility of states for certain internationally wrongful cyber operations.
6. Protection of rights: Helps ensure that investigations respect applicable human rights and legal safeguards.
7. Incident coordination: Supports cooperation in responding to cyberattacks affecting multiple countries.

Example:

An attacker in Country A targets a bank in Country B using a server in Country C. Investigators may need cooperation among all three countries to identify the attacker, preserve evidence and pursue legal action.

Conclusion: International law and cooperation are essential because cybercrime can cross national borders within seconds.

Q3. Explain the responsibilities of the State in maintaining security within cyberspace.

Answer:

The State means the government and its authorized institutions. It has an important role in maintaining a safe, stable and secure cyberspace.

Responsibilities:

1. Formulate laws and policies: Establish legal rules to prevent and

address cybercrime.

2. Protect critical infrastructure: Safeguard important sectors such as energy, banking, transport and communication.

3. Establish cyber agencies: Create institutions for incident response, coordination and threat information.

4. Investigate cybercrime: Support law enforcement and prosecution through lawful procedures.

5. Build national capability: Train cybersecurity professionals and improve technical infrastructure.

6. Public awareness: Educate citizens and organizations about online safety.

7. International cooperation: Work with other countries to address cross-border cyber threats.

8. Protect citizens' rights: Balance cybersecurity measures with privacy, freedom and due process.

9. Develop standards: Encourage secure practices in government departments and public services.

10. Prepare for emergencies: Develop plans to respond to major cyber incidents.

Example:

A government cyber incident response team coordinates with affected organizations during a large-scale cyberattack and provides guidance to limit its impact.

Conclusion: The State must create a secure cyberspace through laws, institutions, infrastructure protection, public awareness and international cooperation.

Q4. Discuss the role of the Private Sector in safeguarding cyberspace.

Answer:

The Private Sector includes companies and organizations that develop, own or operate digital services, networks, software and infrastructure. It plays a major role in cybersecurity because much of cyberspace is operated by private organizations.

Roles:

1. Protect systems and data: Implement firewalls, encryption, access

controls and monitoring.

2. Secure product development: Follow secure coding and testing practices.

3. Incident response: Detect, contain and recover from cyber incidents.

4. Share threat information: Cooperate with government agencies and industry groups.

5. Train employees: Conduct cybersecurity awareness and skill-development programmes.

6. Protect customer information: Use appropriate data-protection and privacy measures.

7. Conduct audits: Assess security controls and fix vulnerabilities.

8. Support critical infrastructure: Protect services such as telecom, cloud, banking and energy systems.

9. Follow standards and regulations: Adopt applicable cybersecurity requirements.

10. Invest in research: Develop improved security technologies and services.

Example:

A cloud service provider protects customer data through encryption, access restrictions, monitoring, backups and incident-response plans.

Conclusion: Government provides laws and coordination, while the private sector implements many of the technical and operational measures needed to safeguard cyberspace.

Q5. Describe the need for Cyber Security Standards in organizations.

Answer:

Cyber Security Standards are documented guidelines and best practices that help organizations establish, implement and maintain effective information security controls.

Need and importance:

1. Consistency: Provide a common approach to security across departments.

2. Risk management: Help identify, assess and treat information security risks.

3. Data protection: Guide organizations in protecting confidential and

sensitive information.

4. Compliance: Help meet legal, regulatory and contractual requirements.

5. Incident management: Provide structured methods for detecting and responding to incidents.

6. Business continuity: Encourage backups, recovery and continuity planning.

7. Customer trust: Demonstrate commitment to protecting information.

8. Continuous improvement: Require regular review and improvement of security controls.

9. Third-party management: Help assess security practices of suppliers and service providers.

Example – ISO/IEC 27001:

An organization can establish an Information Security Management System (ISMS), assess risks, select suitable controls, monitor performance and continually improve its security practices.

Conclusion: Cybersecurity standards help organizations apply systematic, measurable and repeatable security practices.

Q6. Explain the characteristics of the Indian Cyberspace environment.

Answer:

The Indian cyberspace environment includes digital networks, internet services, government platforms, businesses, citizens and the legal and technical systems that support digital activities in India.

Main characteristics:

1. Large and diverse user base: People from different regions, languages and backgrounds use digital services.

2. Rapid digital growth: Online banking, e-commerce, digital payments and government services are widely used.

3. Mobile-first access: A large number of users access the internet through smartphones.

4. Expanding digital infrastructure: Telecom networks, data centres, cloud services and digital platforms continue to develop.

5. Growing cyber threat exposure: Phishing, online fraud, malware, identity theft and data breaches are important concerns.

6. Critical information infrastructure: Sectors such as power, banking,

transport and communication depend on digital systems.

7. Diverse digital literacy: Users have different levels of awareness about online safety and privacy.

8. Multilingual environment: Security awareness and services need to support multiple languages.

9. Regulatory framework: Cybersecurity is addressed through laws, policies, rules and institutions.

10. Public–private cooperation: Government, industry and citizens all contribute to cybersecurity.

Conclusion: India's cyberspace is large, diverse and rapidly developing, so it requires inclusive digital access, strong security practices and coordinated governance.

Q7. Summarize the key objectives of the National Cyber Security Policy (NCSP) 2013.

Answer:

The National Cyber Security Policy (NCSP) 2013 is an Indian policy framework intended to create a secure and resilient cyberspace for citizens, businesses and government.

Key objectives include:

1. Create a secure cyber ecosystem through appropriate policies and institutional structures.

2. Protect information and information infrastructure.

3. Reduce vulnerabilities and improve the security posture of systems.

4. Strengthen capabilities to prevent, detect and respond to cyber threats.

5. Protect critical information infrastructure.

6. Develop a skilled cybersecurity workforce.

7. Encourage research, development and innovation in cybersecurity.

8. Promote public–private partnership and information sharing.

9. Increase awareness about cybersecurity among users and organizations.

10. Support effective legal and regulatory frameworks.

11. Encourage adoption of security standards and best practices.

12. Improve national and international cooperation.

Conclusion: NCSP 2013 aims to build a secure, trusted and resilient cyber environment by combining technology, people, policies and cooperation.

Q8. Explain the significance of the Indian Evidence Act in investigating cybercrime cases.

Answer:

The Indian Evidence Act, 1872 historically provided rules about the relevance and admissibility of evidence in Indian courts, including provisions dealing with electronic records.

Significance in cybercrime investigations:

1. Electronic evidence: Provides a legal framework for considering electronic records as evidence.
2. Admissibility: Addresses conditions under which electronic records may be admitted in legal proceedings.
3. Certification: Section 65B historically prescribed requirements for certificates relating to certain electronic records.
4. Reliability: Encourages proper collection, preservation and documentation of electronic evidence.
5. Authenticity: Helps establish the source and integrity of electronic records.
6. Court proceedings: Provides a basis for presenting digital evidence in criminal and civil cases.

Examples of electronic evidence:

- Emails and chat messages.
- CCTV recordings.
- Computer files and logs.
- Digital photographs and videos.
- Call detail records and electronic documents.

Important legal update:

The Bharatiya Sakshya Adhiniyam, 2023 replaced the Indian Evidence Act, 1872 from 1 July 2024. For an exam question using the older term, explain the historical role of the Indian Evidence Act and mention the current replacement if relevant.

Conclusion: Evidence law is important because digital material must be collected, authenticated and presented according to legal requirements.

Q9. Describe the relevance of the IPC and Criminal Procedure Code (CrPC) to cyber offences.

Answer:

The Indian Penal Code (IPC) and the Code of Criminal Procedure (CrPC) historically formed important parts of India's criminal justice framework and were relevant to certain cyber offences and their investigation.

Relevance of IPC:

1. Defines criminal offences and punishments.
2. Could apply to cheating, fraud, forgery, criminal intimidation and other offences committed using computers or networks.
3. Helped address traditional crimes carried out through digital means.
4. Could be applied along with special cyber laws where appropriate.

Relevance of CrPC:

1. Provides procedures for investigation of criminal offences.
2. Guides police processes such as recording information, investigation and evidence collection.
3. Provides procedures relating to search, seizure, arrest and court proceedings.
4. Helps regulate how criminal cases move through the justice system.

Example:

If a person uses a fake online identity to cheat victims, relevant cheating provisions and procedural rules may be applied along with applicable provisions of the Information Technology Act.

Current legal note:

The Bharatiya Nyaya Sanhita, 2023 replaced the IPC, and the Bharatiya Nagarik Suraksha Sanhita, 2023 replaced the CrPC from 1 July 2024. The question bank uses the older names, so retain them when writing the historical syllabus answer and mention the replacements where appropriate.

Conclusion: Substantive criminal law defines offences and punishments, while criminal procedure law governs investigation and trial procedures.

Q10. Explain the main purpose of the Electronic Communication Privacy Act (ECPA).

Answer:

The Electronic Communications Privacy Act (ECPA) is a United States federal law enacted in 1986. It extends privacy protections to certain electronic communications and addresses unauthorized interception, access and disclosure.

Main purposes:

1. Protect electronic communications: Provides legal protections for certain wire, oral and electronic communications.
2. Restrict unauthorized interception: Regulates interception of communications.
3. Protect stored communications: Sets rules for government access to certain stored electronic communications.
4. Regulate disclosure: Places limits on disclosure of certain communication contents and records.
5. Define lawful access: Establishes legal procedures and exceptions for authorized access.
6. Support privacy: Seeks to balance communication privacy with legitimate law-enforcement needs.

Examples:

- Unauthorized interception of certain private electronic messages may raise ECPA issues.
- Lawful government access to stored communications must follow applicable legal procedures.

Important note:

ECPA is a United States law, not an Indian cyber law. Its relevance in the question bank is to understand electronic communication privacy in an international legal context.

Conclusion: ECPA aims to protect the privacy of electronic communications and regulate when they may be intercepted, accessed or disclosed.

Q11. Discuss the term Legal Policies in the context of cyber security.

Answer:

Legal Policies in cybersecurity are laws, regulations, rules and official guidelines that define rights, duties and responsibilities relating to the use, protection and governance of digital systems and information.

Objectives:

1. Prevent cybercrime: Define unlawful digital activities and legal consequences.
2. Protect privacy: Set rules for collection, use and disclosure of personal information.

3. Secure data: Establish requirements for handling sensitive information.
4. Define responsibilities: Clarify duties of organizations, service providers and users.
5. Support investigations: Provide legal procedures for collecting and presenting digital evidence.
6. Encourage compliance: Require organizations to follow relevant standards and regulations.
7. Protect critical infrastructure: Support safeguards for essential digital services.
8. Promote trust: Provide legal certainty for digital transactions and communication.

Examples in India:

- Information Technology Act, 2000.
- Applicable rules and regulations relating to information technology and data protection.
- Cybersecurity policies and standards issued by competent authorities.

Conclusion: Legal policies create the legal foundation for secure, responsible and accountable use of cyberspace.

Q12. Differentiate between national and international cyber security regulations with examples.

Answer:

National cybersecurity regulations are laws and rules created by one country and generally apply within its jurisdiction. International cybersecurity regulations, agreements and frameworks support cooperation and common approaches between countries.

National regulations:

- Created by a country's legislature or authorized authorities.
- Apply within the country's legal jurisdiction.
- Address domestic cybercrime, data protection and infrastructure security.
- Enforced by national agencies and courts.
- Example: India's Information Technology Act, 2000.

International regulations and frameworks:

- Involve cooperation, agreements or common standards among countries.

- Help address cross-border cybercrime and cyber threats.
- Support information sharing, evidence cooperation and common approaches.
- Implementation may depend on national law and applicable treaties.
- Examples: Budapest Convention on Cybercrime and international standards such as ISO/IEC 27001 (a standard, not a treaty).

Differences:

1. Scope: National rules apply domestically; international instruments involve cross-border cooperation.
2. Authority: National laws are enacted by a country's institutions; international agreements require participation and implementation by states.
3. Enforcement: National laws are enforced by domestic authorities; international cooperation generally works through participating states and their legal systems.
4. Purpose: National rules govern local conduct; international instruments help coordinate responses across borders.

Conclusion: National and international approaches complement each other because cyber threats can affect systems in several countries.

Q13. Analyze how international cyber laws help resolve cybercrimes involving multiple countries.

Answer:

International cyber laws and cooperation frameworks help countries investigate and respond to cybercrimes that involve victims, suspects, servers or evidence in different jurisdictions.

How they help:

1. Legal cooperation: Countries can request assistance through formal legal channels.
2. Evidence collection: Procedures may help preserve and obtain electronic evidence located abroad.
3. Extradition: Applicable agreements and national laws may allow suspects to be transferred for prosecution.
4. Common offence definitions: Shared legal approaches can reduce gaps between national laws.
5. Information sharing: Authorities can exchange relevant threat and

investigation information.

6. Coordinated investigations: Multiple countries can coordinate searches, seizures and investigations within their laws.

7. Capacity building: International cooperation can support training and technical assistance.

Example:

A phishing group operates from Country A, hosts a fake website in Country B and targets victims in Country C. Authorities may coordinate to preserve server records, identify suspects and pursue lawful prosecution.

Challenges:

- Different national laws.
- Different privacy and evidence rules.
- Delays in cross-border requests.
- Encryption and anonymous services.
- Jurisdiction and sovereignty issues.

Conclusion: International cooperation is essential for effective cybercrime investigations because digital evidence and offenders may be spread across several countries.

Q14. Examine a cyber-attack scenario and determine the role of the State versus the Private Sector in mitigating the impact.

Answer:

Scenario:

A ransomware attack affects a private hospital network. Patient records and appointment systems become unavailable, disrupting healthcare services.

Role of the State:

1. Coordinate national cyber incident response through relevant agencies.
2. Provide technical advisories and threat information.
3. Support law enforcement investigation.
4. Protect public safety and critical services.
5. Coordinate with other government departments and international partners if required.
6. Issue applicable regulatory directions and guidance.
7. Support recovery coordination where necessary.

Role of the Private Sector / Hospital:

1. Activate its incident-response plan.
2. Isolate affected systems to limit spread.
3. Preserve logs, malware samples and other evidence.
4. Restore services from clean and verified backups.
5. Protect patient data and communicate with affected stakeholders as required.
6. Engage cybersecurity experts and service providers.
7. Patch vulnerabilities and strengthen access controls.
8. Review the incident and improve security.

Shared responsibilities:

- Exchange relevant threat information.
- Coordinate response and recovery.
- Protect sensitive information.
- Review lessons learned and prevent recurrence.

Conclusion: The State provides legal authority, coordination and national-level support, while the affected organization is responsible for securing its systems, responding operationally and restoring services.

Q15. Apply cyber security standards (such as ISO/IEC 27001) to ensure secure data handling in an organization.

Answer:

ISO/IEC 27001 is an international standard that specifies requirements for establishing, implementing, maintaining and continually improving an Information Security Management System (ISMS).

Application in an organization:

1. Define scope: Identify departments, systems, locations and information covered by the ISMS.
2. Identify assets: List data, applications, devices, networks and services.
3. Assess risks: Identify threats, vulnerabilities and possible business impacts.
4. Treat risks: Select suitable security controls and document decisions.
5. Establish policies: Create rules for access, data handling, passwords and incident reporting.
6. Control access: Apply least privilege, role-based permissions and

authentication.

7. Protect data: Use encryption, secure storage, backups and safe disposal procedures.

8. Manage suppliers: Assess third-party security risks and responsibilities.

9. Train employees: Conduct awareness programmes and role-based training.

10. Monitor and audit: Review controls, logs, incidents and compliance.

11. Improve continuously: Correct nonconformities and update the ISMS as risks change.

Example:

A company handling customer records classifies data, limits access to authorized staff, encrypts sensitive information, maintains backups and audits access logs.

Conclusion: ISO/IEC 27001 provides a systematic risk-based approach to protecting organizational information and improving security continuously.

Q16. Evaluate the effectiveness of the National Cyber Security Policy 2013 in strengthening cybersecurity in India.

Answer:

The National Cyber Security Policy (NCSP) 2013 was designed to create a secure and resilient cyberspace in India through policy, institutional development, skills, awareness and cooperation.

Positive contributions:

1. Established a national policy direction for cybersecurity.
2. Highlighted protection of information and critical infrastructure.
3. Encouraged development of cybersecurity skills and workforce.
4. Promoted public-private partnership.
5. Supported awareness and capacity building.
6. Emphasized research, innovation and security standards.
7. Recognized the need for incident response and cooperation.

Challenges and limitations:

1. Cyber threats continue to evolve rapidly.
2. Implementation across different organizations can be uneven.
3. Shortage of skilled professionals remains a challenge.
4. Small organizations may have limited resources.
5. Coordination between stakeholders can be complex.

6. New technologies create security and governance challenges.
7. Policy goals require regular review and measurable implementation.

Evaluation:

The policy provided an important strategic foundation and helped increase attention to cybersecurity. Its effectiveness depends on implementation, institutional capacity, investment, public awareness and adaptation to emerging threats.

Conclusion: NCSP 2013 was an important step in strengthening India's cybersecurity framework, but continuous updates and effective implementation are necessary to address modern cyber risks.

Q17. Analyze the role of the Indian Evidence Act in presenting digital evidence in court.

Answer:

The Indian Evidence Act, 1872 historically provided rules for presenting evidence in Indian courts, including electronic records.

Role in presenting digital evidence:

1. Recognition of electronic records: Recognized electronic records as a form of evidence subject to legal requirements.
2. Admissibility: Set conditions for admitting certain electronic records.
3. Section 65B: Provided requirements for certificates relating to computer output in specified circumstances.
4. Authenticity: Helped establish the source and reliability of electronic records.
5. Integrity: Encouraged proper preservation and documentation of digital material.
6. Court evaluation: Provided a legal framework for courts to assess electronic evidence.

Examples:

- Emails.
- CCTV recordings.
- Digital photographs and videos.
- Computer logs.
- Electronic documents and messages.

Good forensic practices:

- Maintain chain of custody.

- Create forensic images where appropriate.
- Record hash values.
- Preserve original devices and files.
- Document tools and procedures.
- Obtain required certificates and follow applicable legal rules.

Current legal note:

The Bharatiya Sakshya Adhiniyam, 2023 replaced the Indian Evidence Act from 1 July 2024. The question bank uses the older law's name, so mention the historical framework and current replacement where appropriate.

Conclusion: Evidence law is essential to ensure digital material is collected, authenticated and presented in a legally acceptable manner.

Q18. Compare how the IPC and CrPC deal with cyber offences and suggest improvements.

Answer:

The Indian Penal Code (IPC) and Code of Criminal Procedure (CrPC) historically had different but complementary roles in dealing with cyber offences.

IPC:

- Defines criminal offences and punishments.
- Could apply to cheating, fraud, forgery, intimidation and related conduct committed using digital systems.
- Deals with the substantive criminal liability of an offender.

CrPC:

- Provides procedures for investigation and prosecution.
- Guides police investigation, search, seizure and related processes.
- Regulates how criminal cases proceed through courts.

Comparison:

1. Main role: IPC defines offences; CrPC defines criminal procedure.
2. Focus: IPC concerns what conduct is criminal; CrPC concerns how cases are investigated and tried.
3. Outcome: IPC provides punishments; CrPC provides procedural safeguards and processes.

Suggested improvements:

- Update procedures to address cloud and cross-border evidence.
- Improve training for police, prosecutors and judges.
- Strengthen digital evidence preservation and forensic capacity.
- Improve lawful and timely cooperation with service providers.
- Protect privacy and fundamental rights during investigations.
- Improve coordination between cybercrime agencies.

Current legal note:

The Bharatiya Nyaya Sanhita, 2023 replaced the IPC, and the Bharatiya Nagarik Suraksha Sanhita, 2023 replaced the CrPC from 1 July 2024.

Conclusion: Criminal law defines offences and punishments, while procedural law ensures investigations and trials follow lawful methods. Both are necessary for addressing cybercrime.

Q19. Investigate a case of email privacy violation under the Electronic Communication Privacy Act (ECPA) and suggest applicable legal actions.

Answer:

Scenario:

An employee's private email communications are accessed or intercepted without proper authorization by another person. The affected individual alleges a violation of email privacy.

ECPA context:

The Electronic Communications Privacy Act is a United States federal law. It addresses unauthorized interception of certain electronic communications and access to certain stored communications.

Investigation steps:

1. Preserve relevant emails, headers, access logs and account notifications.
2. Record dates, times, affected accounts and suspected access.
3. Secure the account by changing credentials and enabling MFA.
4. Review login history, forwarding rules and connected applications.
5. Preserve relevant service-provider records through lawful procedures.
6. Identify whether communications were intercepted in transit or accessed while stored.
7. Determine who accessed the account and whether authorization existed.
8. Consult qualified legal counsel regarding applicable ECPA provisions and

exceptions.

Possible legal actions:

- Report suspected unauthorized access to appropriate law-enforcement authorities.
- Seek preservation of relevant electronic records through lawful channels.
- Consider civil remedies where available under applicable law.
- Notify affected organizations or service providers as appropriate.

Important note:

ECPA is a United States law. If the incident occurs in India, Indian laws and applicable rules must also be considered; ECPA does not automatically govern an incident in India.

Conclusion: Email privacy investigations require preservation of electronic evidence, identification of the type of access and application of the correct jurisdiction's law.

Q20. Examine the challenges faced by India in regulating cyberspace and propose solutions.

Answer:

India faces several challenges in regulating cyberspace because digital technologies, cyber threats and online services develop rapidly and often cross national borders.

Challenges:

1. Rapid technological change: Laws and policies may struggle to keep pace with new technologies.
2. Cross-border jurisdiction: Offenders and digital evidence may be located outside India.
3. Increasing cybercrime: Phishing, ransomware, online fraud and identity theft continue to pose risks.
4. Digital literacy gaps: Users have different levels of awareness about online safety.
5. Shortage of skilled professionals: More trained cybersecurity experts are needed.
6. Privacy concerns: Security measures must respect privacy and other legal rights.
7. Critical infrastructure risks: Essential services depend on

interconnected digital systems.

8. Coordination issues: Multiple agencies and private organizations may need to work together.

9. Small-business limitations: Smaller organizations may lack funds and expertise for strong security.

10. Evidence challenges: Encryption, cloud storage and deleted data can complicate investigations.

Suggested solutions:

- Regularly review and update cyber laws and policies.
- Strengthen cybercrime investigation and digital forensic capabilities.
- Improve international cooperation and evidence-sharing procedures.
- Promote cybersecurity education in schools, colleges and workplaces.
- Develop skilled professionals through training and research.
- Encourage public-private partnerships and threat information sharing.
- Improve protection of critical information infrastructure.
- Support affordable security standards for small organizations.
- Ensure cybersecurity measures respect privacy and due process.
- Conduct regular audits, exercises and incident-response drills.

Conclusion: India needs a balanced approach combining effective laws, skilled people, strong technology, public awareness and international cooperation.

Q21. Apply cyber security regulatory frameworks to safeguard critical information infrastructure in India.

Answer:

Critical Information Infrastructure (CII) refers to computer resources whose destruction or incapacitation may have a debilitating impact on national security, economy, public health or safety.

Regulatory and institutional approach in India:

1. Information Technology Act, 2000: Provides the legal framework for electronic records, cyber offences and protection of certain critical systems.

2. Section 70 of the IT Act: Provides for declaration of certain computer resources as protected systems.

3. NCIIPC: The National Critical Information Infrastructure Protection Centre is responsible for protecting critical information infrastructure.

4. CERT-In: Supports cybersecurity incident response, alerts and coordination under its mandate.
5. NCSP 2013: Provides policy direction for creating a secure and resilient cyber ecosystem.
6. Security standards: Organizations can adopt standards such as ISO/IEC 27001 for information security management.

Application steps:
1. Identify critical assets and essential services.
2. Conduct risk and vulnerability assessments.
3. Classify information according to sensitivity.
4. Apply strong access control and multi-factor authentication.
5. Segment networks and protect important systems.
6. Encrypt sensitive information and maintain secure backups.
7. Monitor systems and maintain incident logs.
8. Prepare incident-response and disaster-recovery plans.
9. Conduct audits, testing and employee training.
10. Coordinate with relevant national authorities and report incidents as required.

Example:
A power-sector organization identifies its control systems as critical assets, restricts access, segments operational networks, monitors activity and maintains tested recovery plans.

Conclusion: Protecting CII requires legal compliance, strong technical controls, continuous monitoring and coordination between government and infrastructure operators.

Q22. Assess how legal policies influence organizational adoption of cybersecurity practices.

Answer:
Legal policies influence how organizations manage, protect and use information systems. They establish duties and requirements that encourage organizations to adopt cybersecurity practices.

Influence on organizations:
1. Security governance: Organizations create policies, roles and responsibilities.
2. Risk assessment: They identify threats and vulnerabilities to meet legal

and regulatory requirements.

3. Data protection: They apply controls for collecting, storing, processing and sharing information.

4. Access control: They restrict access to authorized users.

5. Incident response: They create procedures for reporting, investigating and responding to incidents.

6. Employee awareness: They train employees on safe digital practices.

7. Audits and compliance: They review controls and maintain required records.

8. Vendor management: They assess third-party security risks.

9. Investment: They allocate funds for security tools, staff and infrastructure.

10. Accountability: They define responsibilities and consequences for non-compliance.

Example:

A company handling customer information adopts access controls, encryption, retention procedures, incident reporting and security audits to meet applicable legal and regulatory requirements.

Positive effects:

- Better protection of customer information.
- Improved trust and accountability.
- More consistent security procedures.
- Reduced legal and operational risk.

Challenges:

- Compliance costs.
- Complex and changing requirements.
- Shortage of skilled staff.
- Difficulty applying controls to legacy systems.

Conclusion: Legal policies encourage organizations to make cybersecurity a formal management responsibility rather than treating it only as a technical issue.